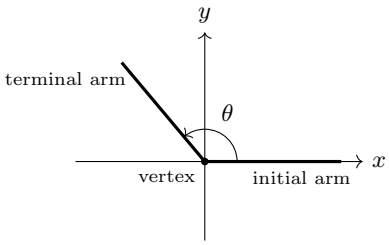
Lesson 1: Angles in Standard Position and Reference Angles

Placing rotation angles on the coordinate plane, naming their quadrants, and finding the acute angle each terminal arm makes with the x-axis.

1. Angles in Standard Position

An angle is in **standard position** when its **vertex** is at the origin and its **initial arm** lies along the positive x-axis. The angle is made by rotating a second arm, the **terminal arm**, away from the initial arm. The amount of turning is the **rotation angle**, θ.

- A **counterclockwise** rotation gives a **positive** angle. One full turn is 360°.
- A clockwise rotation gives a negative angle, and a rotation of more than 360° goes past a full turn. Either way the terminal arm lands on the same arm as some angle from 0° to 360°. In this course, $0° \le \theta \le 360°$.

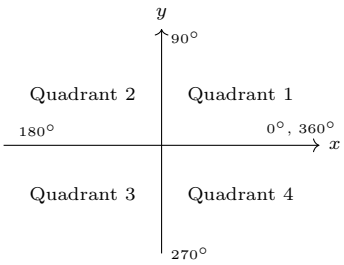


2. Quadrants and Quadrantal Angles

The axes split the plane into four **quadrants**, numbered counterclockwise from the positive x-axis. An angle whose terminal arm lies *on an axis* (0°, 90°, 180°, 270°, 360°) is a **quadrantal angle**; it is not in any quadrant.

Quadrant 1: $0° < \theta < 90°$	Quadrant 2: $90° < \theta < 180°$
Quadrant 3: $180° < \theta < 270°$	Quadrant 4: $270° < \theta < 360°$

To sketch an angle, start on the positive x-axis, turn counterclockwise, and stop in the right quadrant: halfway through a quadrant is 45° past the axis.



Example 1

Sketch each angle in standard position and state where its terminal arm lies.

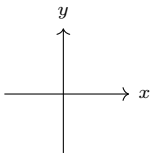
(a) 160° (b) 245° (c) 335° (d) 90°



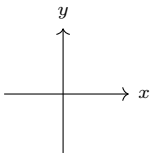
Example 2

Sketch each rotation. Then state the angle from 0° to 360° that has the same terminal arm.

(a) a clockwise rotation of 135° (that is, −135°)



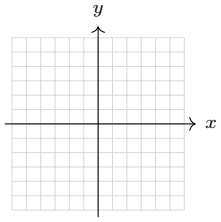
(b) a counterclockwise rotation of 420°



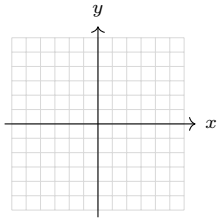
Example 3

Each point lies on the terminal arm of an angle θ in standard position. Plot the point, draw θ, and state its quadrant.

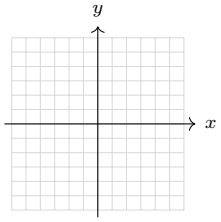
(a) $A(-6, 2)$



(b) $B(1, -3)$

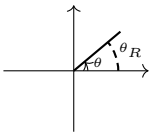


(c) $C(-2, -5)$

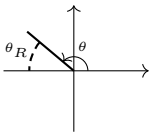


3. Reference Angles

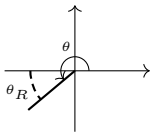
The **reference angle**, θ_R, is the acute angle between the terminal arm and the x**-axis**. It is always positive and between 0° and 90°, and it is always measured to the x-axis, *never* to the y-axis. In each quadrant, look at how far the terminal arm is from the nearest part of the x-axis (0°, 180° or 360°).



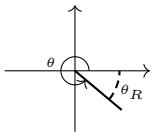
Quadrant 1
$\theta_R = \theta$



Quadrant 2
$\theta_R = 180° - \theta$



Quadrant 3
$\theta_R = \theta - 180°$



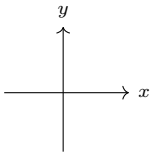
Quadrant 4
$\theta_R = 360° - \theta$

For a quadrantal angle, the arm is on an axis: $\theta_R = 0°$ for 0°, 180°, 360° (on the x-axis) and $\theta_R = 90°$ for 90°, 270°.

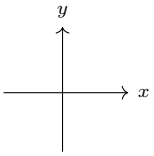
Example 4

Sketch each angle and its reference angle. Determine θ_R.

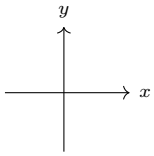
(a) 205°



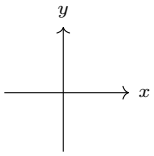
(b) 118°



(c) 289°



(d) 47°



Example 5

Work backwards: determine the rotation angle θ, $0° \le \theta \le 360°$, from the reference angle and where the terminal arm lies.

Reference angle	Terminal arm	Sketch	Rotation angle θ
35°	quadrant 3		
72°	quadrant 2		
16°	quadrant 4		
50°	quadrant 1		
0°	negative x-axis		
90°	negative y-axis		

4. Four Angles with the Same Reference Angle

Turning the same acute angle away from the x-axis in each quadrant gives four angles with one reference angle. Rearranging the formulas in the box above:

For a reference angle θ_R, the angles from 0° to 360° are θ_R (Q1), $180° - \theta_R$ (Q2), $180° + \theta_R$ (Q3), $360° - \theta_R$ (Q4).

Example 6

Determine the three other angles from 0° to 360° that have the same reference angle as the given angle.

(a) 128°

(b) 341°

5. Reflecting a Point in the Axes

Reflecting the terminal arm in an axis keeps the reference angle and moves the arm to another quadrant. If $P(a, b)$ with $a, b > 0$ is on the terminal arm of θ in quadrant 1:

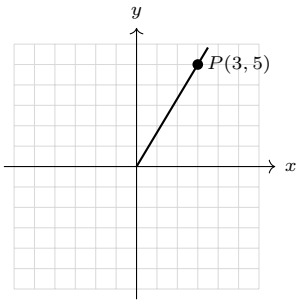
reflect in the y-axis:	$(-a, b)$	on the arm of $180° - \theta$ (Q2)
reflect in both axes (through the origin):	$(-a, -b)$	on the arm of $180° + \theta$ (Q3)
reflect in the x-axis:	$(a, -b)$	on the arm of $360° - \theta$ (Q4)

Example 7

The point $P(3, 5)$ is on the terminal arm of an angle of 59° (to the nearest degree).

(a) Plot Q, the reflection of P in the y-axis; R, the reflection of P in both axes; and S, the reflection of P in the x-axis. State their coordinates.

(b) State the rotation angle whose terminal arm passes through each of Q, R and S.

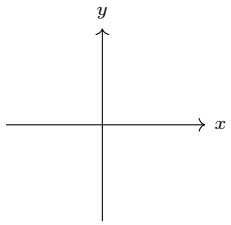


6. Common Mistakes with Reference Angles

Most wrong reference angles come from measuring to the y-axis, or from using the formula for the wrong quadrant. A quick sketch prevents both: the answer must be the gap between the arm and the x-axis, and it must be between $0°$ and $90°$.

Example 8

Three students found the reference angle for $251°$. Ana said $19°$, Ben said $109°$ and Cole said $71°$. Who is correct? Explain each mistake with a sketch.



Lesson 2: Trigonometric Ratios from a Point on the Terminal Arm

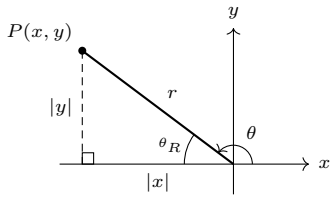
Using the coordinates of a point on the terminal arm to find r, the exact values of $\sin\theta$, $\cos\theta$ and $\tan\theta$, and the angle θ itself.

1. The Distance r from the Origin

Let $P(x, y)$ be any point on the terminal arm of θ, other than the origin. Dropping a perpendicular from P to the x-axis makes a right triangle with legs of length $|x|$ and $|y|$. Its hypotenuse is the distance r from the origin to P.

$x^2 + y^2 = r^2$, so $r = \sqrt{x^2 + y^2}$ (r is a distance: always $r > 0$)

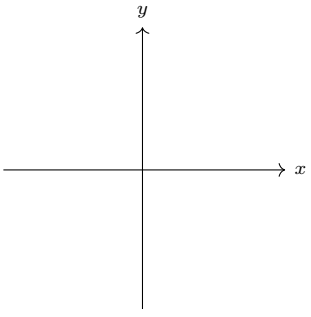
The triangle is the **reference triangle**: its angle at the origin is the reference angle θ_R. Leave r as an exact radical in simplest form when it is not a whole number.

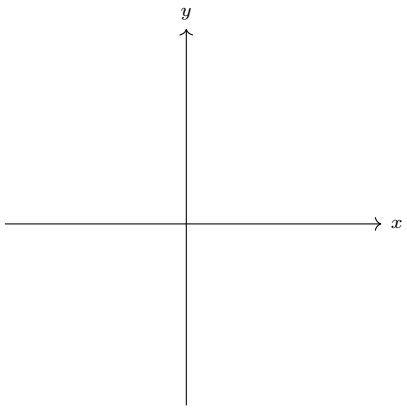


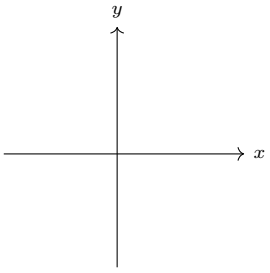
Example 1

Sketch θ and determine the exact distance r from the origin to the point on its terminal arm.

(a) $P(-20, 21)$ (b) $Q(4, -6)$ (c) $R(-3, -3)$







2. The Ratios in Terms of x, y and r

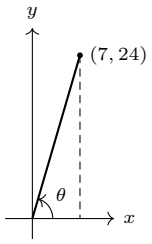
In Math 10C, $\sin\theta = \dfrac{\text{opposite}}{\text{hypotenuse}}$, $\cos\theta = \dfrac{\text{adjacent}}{\text{hypotenuse}}$ and $\tan\theta = \dfrac{\text{opposite}}{\text{adjacent}}$ (SOH CAH TOA) for an acute angle in a right triangle. For an angle in standard position, the ratios are defined with the coordinates of $P(x, y)$:

$$\sin\theta = \frac{y}{r} \qquad \cos\theta = \frac{x}{r} \qquad \tan\theta = \frac{y}{x} \quad (x \neq 0)$$

In quadrant 1 these are exactly SOH CAH TOA. In other quadrants x or y is negative, so a ratio can be **negative**. Since $r > 0$, the sign of $\sin\theta$ is the sign of y, and the sign of $\cos\theta$ is the sign of x. Write exact answers with a rational denominator (Unit 2).

Example 2

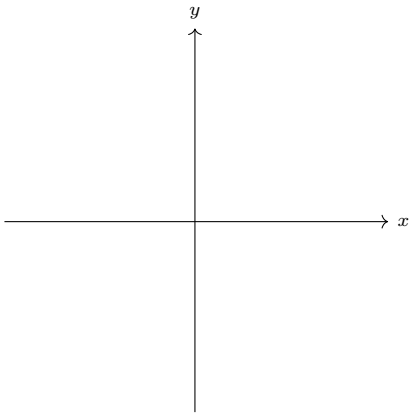
The point $(7, 24)$ lies on the terminal arm of θ. Determine r and the exact values of $\sin\theta$, $\cos\theta$ and $\tan\theta$.



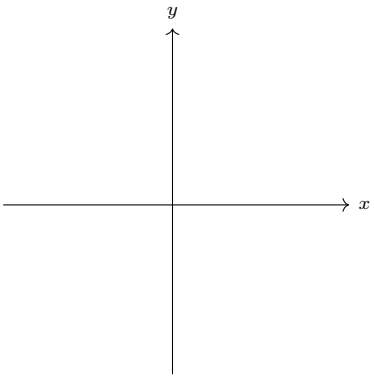
Example 3

Each point lies on the terminal arm of θ. Sketch θ, then determine the exact values of $\sin\theta$, $\cos\theta$ and $\tan\theta$.

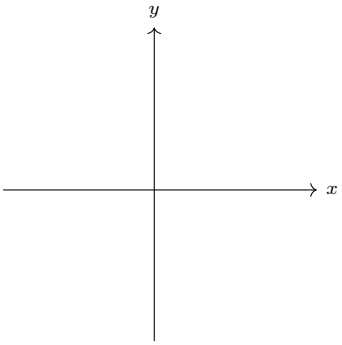
(a) $(-9, 40)$



(b) $(-5, -12)$



(c) $(6, -3)$

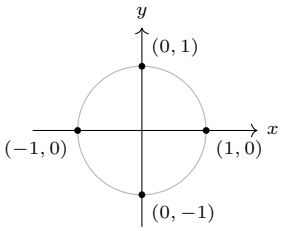


(d) In each part, which ratios are negative? What decides the sign?

3. Quadrantal Angles

For a quadrantal angle, the terminal arm lies on an axis, so one coordinate of P is 0. Using the point with $r = 1$ on each axis makes the values easy to read. When $x = 0$, $\tan\theta = \dfrac{y}{0}$ is **undefined** (we cannot divide by zero).

Example 4



(a) Each point shown is on the terminal arm of a quadrantal angle, with $r = 1$. Complete the table.

θ	$0°$	$90°$	$180°$	$270°$	$360°$
point	$(1, 0)$	$(0, 1)$	$(-1, 0)$	$(0, -1)$	$(1, 0)$
$\sin\theta$					
$\cos\theta$					
$\tan\theta$					

(b) The point $(0, -7)$ is on the terminal arm of θ. State θ and the values of $\sin\theta$, $\cos\theta$ and $\tan\theta$.

4. Same Reference Angle, Same Ratios up to Sign

Reflecting P in the axes keeps the reference triangle the same size, so the ratios keep the same *size*; only the signs of x and y change.

Example 5

The point $P(3, 4)$ is on the terminal arm of $\theta \approx 53.1°$. Reflect P in the y-axis, in both axes, and in the x-axis, and complete the table.

Point	$P(3, 4)$	$Q(,\ )$	$R(,\ )$	$S(,\ )$
Quadrant	1			
r	5			
θ (nearest tenth)	$53.1°$			
$\sin\theta$	$\frac{4}{5}$			
$\cos\theta$	$\frac{3}{5}$			
$\tan\theta$	$\frac{4}{3}$			

Check one column with a calculator in Degree mode, for example $\sin 233.1° \approx -0.80$. What do you notice?

The value of $\sin\theta$, $\cos\theta$ or $\tan\theta$ is the value of the same ratio of θ_R, with a sign decided by the quadrant: positive or negative as y (for sine), x (for cosine) or $\dfrac{y}{x}$ (for tangent) is in that quadrant.

Example 6

Use a sketch to write each ratio as the same ratio of the reference angle, with the correct sign.

(a) $\sin 250°$ (b) $\cos 130°$ (c) $\tan 310°$ (d) $\sin 165°$

5. Finding the Angle θ from a Point

The inverse tangent key gives the reference angle when it is given the *positive* ratio of the legs. Set the calculator to **Degree mode** first (TI-84: [MODE], choose DEGREE).

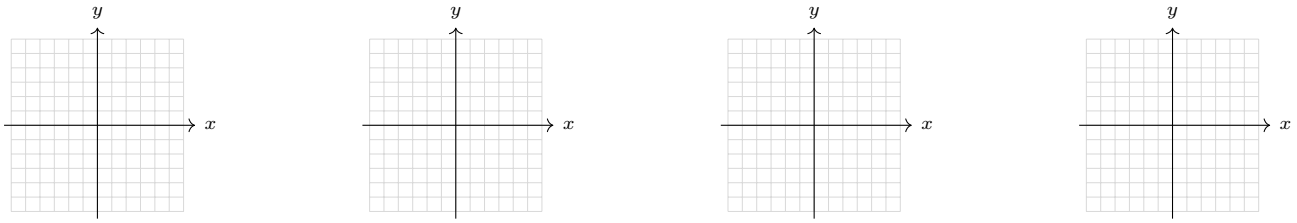
1. Plot $P(x, y)$ and decide its quadrant. (If P is on an axis, θ is 0°, 90°, 180° or 270°: no calculator needed.)
2. Find the reference angle: $\theta_R = \tan^{-1}\left(\dfrac{|y|}{|x|}\right)$ (2nd [TAN^{-1}]).
3. Place θ in its quadrant: Q1 $\theta = \theta_R$; Q2 $\theta = 180° - \theta_R$; Q3 $\theta = 180° + \theta_R$; Q4 $\theta = 360° - \theta_R$.

Entering $\tan^{-1}\left(\dfrac{y}{x}\right)$ with the signs left in gives an angle from $-90°$ to $90°$, which is wrong outside quadrant 1. Using $|x|$ and $|y|$ and a sketch avoids this.

Example 7

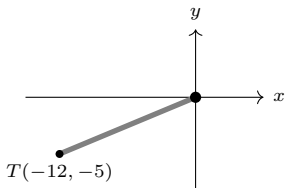
Each point is on the terminal arm of θ, $0° \le \theta < 360°$. Sketch θ and determine it to the nearest tenth of a degree.

(a) $(5, 2)$ (b) $(-4, 5)$ (c) $(-6, -1)$ (d) $(3, -5)$



Example 8

A robotic arm pivots at the origin of a coordinate grid marked in centimetres. The tip of the arm is at $T(-12, -5)$. The rotation angle θ is measured counter-clockwise from the positive x-axis.



(a) How long is the arm?

(b) Determine the exact values of $\sin\theta$, $\cos\theta$ and $\tan\theta$.

(c) Determine θ to the nearest tenth of a degree.

Lesson 3: The CAST Rule: Signs and Solving for Angles

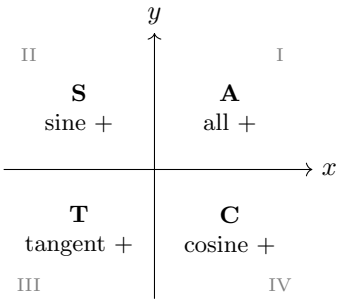
Using the sign of each ratio in each quadrant to find exact ratios from one known ratio, and to solve $\sin\theta = a$, $\cos\theta = a$ *and* $\tan\theta = a$ *for* $0° \le \theta < 360°$.

1. Signs of the Ratios: the CAST Rule

For a point $P(x, y)$ on the terminal arm, $\sin\theta = \frac{y}{r}$, $\cos\theta = \frac{x}{r}$ and $\tan\theta = \frac{y}{x}$, where $r > 0$ always. So the sign of a ratio depends only on the signs of x and y:

- $\sin\theta$ has the sign of y (positive above the x-axis);
- $\cos\theta$ has the sign of x (positive right of the y-axis);
- $\tan\theta$ is positive when x and y have the *same* sign.

The diagram shows which ratios are **positive** in each quadrant. Read the letters counterclockwise from Quadrant IV: **C, A, S, T** — the **CAST rule**. (Memory aid from Quadrant I: *All Students Take Calculus*.)



The ratios not named in a quadrant are negative there. On the axes (quadrantal angles) a ratio is 0, 1, -1 or undefined, as in Lesson 2.

Example 1

(a) Complete the table with + or −.

	I	II	III	IV
$\sin\theta$				
$\cos\theta$				
$\tan\theta$				

(b) In which quadrant(s) does θ terminate if

(i) $\cos\theta > 0$? _____

(ii) $\tan\theta < 0$? _____

(iii) $\sin\theta < 0$ and $\tan\theta > 0$? _____

(c) Without a calculator, state whether each ratio is positive or negative. Name the quadrant first.

(i) $\sin 162°$ (ii) $\cos 251°$ (iii) $\tan 304°$ (iv) $\tan 200°$

2. Writing a Ratio Using Its Reference Angle

The terminal arms of θ and of its reference angle θ_R make the same right triangle with the x-axis, only with different signs on x and y. So a ratio of θ has the same *size* as that ratio of θ_R, and CAST gives the *sign*.

Ratio of θ = ± (same ratio of θ_R), with the sign from CAST for the quadrant of θ.
Example: $150°$ is in Quadrant II with $\theta_R = 30°$, so $\sin 150° = \sin 30°$, $\cos 150° = -\cos 30°$, $\tan 150° = -\tan 30°$.

This also finds the *other* angle from $0°$ to $360°$ with the same ratio: it has the same reference angle and lies in the other quadrant where that ratio has the same sign.

Example 2

Write each ratio as the same ratio of its reference angle, with the correct sign.

(a) $\cos 132°$ (b) $\sin 214°$ (c) $\tan 297°$ (d) $\sin 349°$

(e) Determine the other angle θ, $0° \le \theta < 360°$, with $\cos\theta = \cos 132°$.

(f) Determine the other angle θ, $0° \le \theta < 360°$, with $\tan\theta = \tan 128°$.

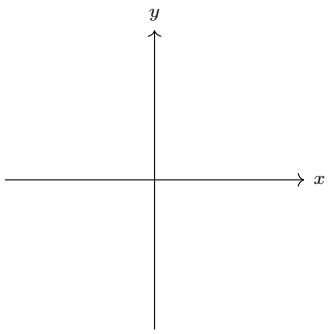
3. Finding the Other Ratios from One Ratio

If one ratio and enough sign information are given, the other two ratios can be found exactly:

1. Use the signs (CAST) to decide the quadrant.
2. Read two of x, y, r from the given ratio, with the signs for that quadrant (r is always positive).
3. Find the third with $x^2 + y^2 = r^2$, and give it the sign for the quadrant.
4. Sketch the angle and its reference triangle, then write the ratios. Rationalize any radical denominator.

Example 3

Given $\cos\theta = -\dfrac{8}{17}$ and $\tan\theta > 0$, determine the exact values of $\sin\theta$ and $\tan\theta$.



Example 4

Determine the exact values of the other two ratios. Rationalize the denominators.

(a) $\tan\theta = \dfrac{2}{5}$ and $\sin\theta < 0$

(b) $\sin\theta = \dfrac{2}{3}$ and $90° < \theta < 180°$

4. Solving $\sin\theta = a$, $\cos\theta = a$ and $\tan\theta = a$

An equation like $\sin\theta = 0.42$ usually has **two** solutions from $0°$ to $360°$, because the ratio has the same sign in two quadrants. Set the calculator to **Degree mode** first (TI-84: MODE, choose DEGREE).

To solve $\sin\theta = a$, $\cos\theta = a$ or $\tan\theta = a$ for $0° \le \theta < 360°$:

1. Use the sign of a and CAST to find the two quadrants.
2. Find the reference angle with the inverse key on the *positive* value: $\theta_R = \sin^{-1}|a|$ (2nd [SIN]), $\cos^{-1}|a|$ or $\tan^{-1}|a|$.
3. Place θ_R in each quadrant: I: θ_R II: $180° - \theta_R$ III: $180° + \theta_R$ IV: $360° - \theta_R$.

Use $|a|$ in step 2 so the calculator always returns an acute angle; $\sin^{-1}(-0.42)$ gives $-24.8°$, which is not in the interval. Keep the full calculator value and round only the final answers.

Example 5

Solve for θ, $0° \leq \theta < 360°$, to the nearest tenth of a degree.

(a) $\sin\theta = 0.42$ (b) $\cos\theta = -0.73$ (c) $\tan\theta = -3.5$

5. Quadrantal Solutions and Equations with No Solution

Since $\sin\theta = \frac{y}{r}$ and $\cos\theta = \frac{x}{r}$ with $|x| \leq r$ and $|y| \leq r$, the values of sine and cosine are always between -1 and 1. So $\sin\theta = a$ and $\cos\theta = a$ have **no solution** when $a > 1$ or $a < -1$. Tangent can be any real number, so $\tan\theta = a$ always has two solutions.

When a is 0, 1 or -1, the solutions are quadrantal angles, and there may be only one. Use the values from Lesson 2:

θ	$0°$	$90°$	$180°$	$270°$
$\sin\theta$	0	1	0	-1
$\cos\theta$	1	0	-1	0
$\tan\theta$	0	undefined	0	undefined

Example 6

Solve for θ, $0° \leq \theta < 360°$.

(a) $\sin\theta = -1$ (b) $\cos\theta = 0$ (c) $\tan\theta = 0$ (d) $\cos\theta = -1.15$

6. Rearranging First, Restricted Intervals and Problems

If the ratio is not alone, isolate it first, exactly as you would isolate x in a linear equation. If the question gives a smaller interval (such as $0° \leq \theta \leq 180°$), find all the solutions, then keep only those in the interval.

Example 7

Solve, to the nearest tenth of a degree.

(a) $5\sin\theta + 2 = 0$, $0° \leq \theta < 360°$ (b) $3(2 - \cos\theta) = 4$, $0° \leq \theta < 360°$ (c) $3\tan\theta + 8 = 0$, $0° \leq \theta \leq 180°$

Example 8

A bicycle pedal turns about its axle. When the crank arm has rotated θ from the horizontal (in standard position), the height of the pedal above the axle is $h = 17\sin\theta$, in centimetres. A negative height means the pedal is below the axle.

(a) For $0° \le \theta < 360°$, at what angles is the pedal 6 cm below the axle? Answer to the nearest tenth of a degree.

(b) Can the pedal ever be 20 cm above the axle? Explain using the equation.

Lesson 4: Exact Values from Special Triangles

Using the 45-45-90 and 30-60-90 triangles, reference angles and CAST to find exact ratios and to solve equations exactly, without a calculator.

1. The Two Special Triangles

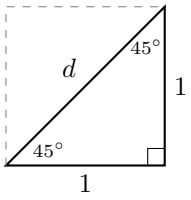
Most ratios are irrational decimals, but the angles 30°, 45° and 60° have **exact values** that come from two triangles.

- Cut a square with side 1 along a diagonal: each half is a right isosceles triangle with angles 45°, 45°, 90°.
- Cut an equilateral triangle with side 2 by an altitude: each half has angles 30°, 60°, 90°, hypotenuse 2 and shortest side 1.

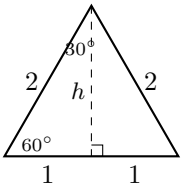
No calculator in this lesson: every answer is exact. Write a radical denominator in rationalized form, as in Unit 2: $\dfrac{1}{\sqrt{2}} = \dfrac{\sqrt{2}}{2}$ and $\dfrac{1}{\sqrt{3}} = \dfrac{\sqrt{3}}{3}$.

Example 1

(a) The square has side 1. Find the diagonal d, then the ratios of 45°.



(b) The equilateral triangle has side 2. Find the altitude h, then the ratios of 30° and 60°.

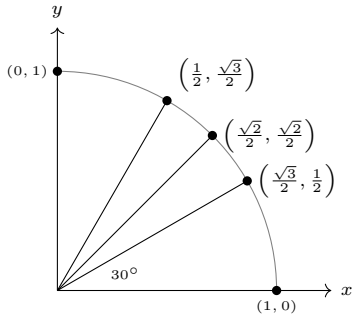


2. The Table of Exact Values

For $0°$ and $90°$ use the points $(1, 0)$ and $(0, 1)$ with $r = 1$ (Lesson 2). The values for $0°$ to $90°$ go in one table that you should know from memory. Patterns to help you remember:

- The sines from $0°$ to $90°$ are $\frac{\sqrt{0}}{2}, \frac{\sqrt{1}}{2}, \frac{\sqrt{2}}{2}, \frac{\sqrt{3}}{2}, \frac{\sqrt{4}}{2}$: they increase from 0 to 1.
- The cosines are the same list in reverse: they decrease from 1 to 0.
- $\tan\theta = \dfrac{\sin\theta}{\cos\theta}$ (since $\dfrac{y/r}{x/r} = \dfrac{y}{x}$), and $\tan 90°$ is undefined because $\cos 90° = 0$.

The picture shows where the values come from: shrink each triangle so its hypotenuse is 1, and its corner lands on a circle of radius 1 at the point $(\cos\theta, \sin\theta)$.



Example 2

Complete the table of exact values. Write each value in rationalized form.

θ	$0°$	$30°$	$45°$	$60°$	$90°$
$\sin\theta$					
$\cos\theta$					
$\tan\theta$					

3. Exact Values in Any Quadrant

Exact value of a ratio of θ: (1) find the quadrant and the reference angle θ_R; (2) if θ_R is $30°$, $45°$ or $60°$, take the value from the table; (3) attach the sign from CAST. For a quadrantal angle use x, y and $r = 1$ directly.

Example 3

Determine the exact value.

(a) $\sin 240°$ (b) $\cos 135°$ (c) $\tan 330°$

(d) $\cos 225°$ (e) $\sin 180°$ (f) $\tan 270°$

4. Exact Expressions

Replace each ratio by its exact value in brackets, then simplify with the order of operations. The notation $\sin^2\theta$ means $(\sin\theta)^2$: find $\sin\theta$ first, then square it.

Example 4

Determine the exact value of each expression.

(a) $\sin^2 60° + \cos 180°$

(b) $2\cos 45° \sin 135°$

(c) $\dfrac{\tan 150°}{1 + \tan^2 210°}$

(d) Is $\sin 60° = 2\sin 30°$? Explain.

5. Solving Equations Exactly

To solve $\sin\theta = a$, $\cos\theta = a$ or $\tan\theta = a$ without a calculator, use the same steps as in Lesson 3, but read the reference angle from the table instead of the inverse key.

(1) Isolate the ratio; rationalize a if needed. (2) Use the sign of a and CAST to choose the quadrants. (3) Find the angle in the table whose ratio is $|a|$: that is θ_R. (4) Place θ_R in each quadrant. If a is 0, 1, -1 or the ratio is undefined, the answers are quadrantal angles.

If the equation has a square, such as $\tan^2\theta = 1$, take the square root of both sides: $\tan\theta = \pm 1$. Then both signs give solutions, so there can be four.

Example 5

Solve for θ, $0° \le \theta < 360°$, without a calculator.

(a) $\sin\theta = -\dfrac{\sqrt{3}}{2}$

(b) $\cos\theta = -\dfrac{1}{\sqrt{2}}$

(c) $\tan\theta = \sqrt{3}$

Example 6

Solve, without a calculator.

(a) $2\sin\theta + 1 = 0$, $0° \le \theta < 360°$

(b) $\sqrt{3}\tan\theta + 1 = 0$, $0° \le \theta \le 180°$

(c) $\tan^2\theta = 1$, $0° \le \theta < 360°$

6. Finding an Exact Angle from a Point or a Ratio

If a point on the terminal arm gives a ratio from the table, the angle can be found exactly: find $\tan\theta = \frac{y}{x}$ (or r and then sine or cosine), match the reference angle, and place it in the quadrant of the point.

Example 7

(a) The point $P(2\sqrt{3}, -2)$ is on the terminal arm of θ. Determine θ exactly, $0° \le \theta < 360°$.

(b) The point $Q(-3, -3\sqrt{3})$ is on the terminal arm of θ. Determine θ and r.

(c) $\sin A = \frac{1}{2}$ and A is not in Quadrant I. Determine A, $0° \le A < 360°$, and the exact values of $\cos A$ and $\tan A$.

(d) A terminal arm rotates counterclockwise from the arm through $(-1, 1)$ to the arm through $(1, -\sqrt{3})$. Through what angle does it rotate?

Lesson 5: The Sine Law

Relating each side of any triangle to the sine of the angle opposite it, then finding unknown sides and angles.

1. Labelling Triangles and a Right-Triangle Review

From now on a triangle is not always a right triangle. Name the vertices with capital letters: in $\triangle ABC$ the angles are $\angle A$, $\angle B$ and $\angle C$, and the side **opposite** each angle (the side it does not touch) gets the matching **lowercase** letter: a is opposite $\angle A$, b is opposite $\angle B$, c is opposite $\angle C$. The three angles always add to 180°. Set the calculator to **Degree mode** (TI-84: MODE, choose DEGREE).

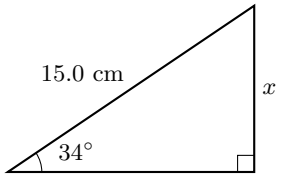
In a right triangle, the primary ratios connect an acute angle θ to two sides.

$$\sin\theta = \frac{\text{opposite}}{\text{hypotenuse}} \qquad \cos\theta = \frac{\text{adjacent}}{\text{hypotenuse}} \qquad \tan\theta = \frac{\text{opposite}}{\text{adjacent}} \qquad \text{(SOH CAH TOA)}$$

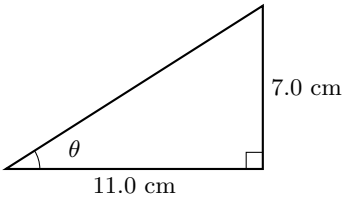
Example 1

Round sides to the nearest tenth and angles to the nearest degree.

(a) Determine x.



(b) Determine θ.

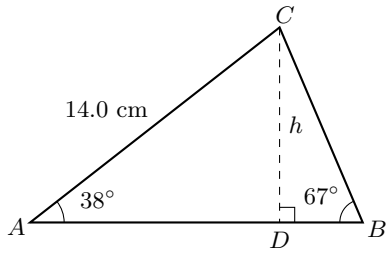


2. Splitting a Triangle into Right Triangles

A triangle with no right angle is an **oblique** triangle; SOH CAH TOA does not apply to it directly. Drawing an **altitude** (a height, perpendicular to a side) splits it into two right triangles that SOH CAH TOA can solve.

Example 2

In $\triangle ABC$, $\angle A = 38°$, $\angle B = 67°$ and $AC = 14.0$ cm. The altitude CD has length h.



(a) Use $\triangle ACD$ to determine h to the nearest hundredth.

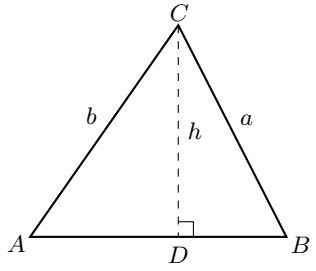
(b) Use $\triangle BCD$ to determine BC to the nearest tenth.

3. The Sine Law

Example 2 wrote the same altitude two ways. Doing this for any triangle gives a rule that skips the altitude.

Example 3

Complete the proof. CD is the altitude from C, with length h.



In $\triangle ACD$: $\sin A =$ _____, so $h =$ _____.

In $\triangle BCD$: $\sin B =$ _____, so $h =$ _____.

Both equal h, so _____ = _____.

Divide both sides by $\sin A \sin B$: _____

An altitude from A gives $\dfrac{b}{\sin B} = \dfrac{c}{\sin C}$ in the same way. If an angle is **obtuse**, the altitude falls outside the triangle, and the proof uses $\sin(180° - \theta) = \sin\theta$: the two angles have the same reference angle and sine is positive in quadrant II (Lesson 3). For example, $\sin 130° = \sin 50° \approx 0.766$. The law holds in every triangle.

The Sine Law. In any $\triangle ABC$: $\dfrac{a}{\sin A} = \dfrac{b}{\sin B} = \dfrac{c}{\sin C}$ or $\dfrac{\sin A}{a} = \dfrac{\sin B}{b} = \dfrac{\sin C}{c}$

Use two of the ratios at a time. One of them must be a **known pair**: an angle and its opposite side. Put the unknown on top: sides on top to find a side, sines on top to find an angle.

Example 4

Use the sine law in the triangle from Example 2 ($\angle A = 38°$, $\angle B = 67°$, $b = 14.0$ cm). Round to the nearest tenth.

(a) Determine $a = BC$. Compare with Example 2.

(b) Determine $c = AB$.

4. Finding a Side

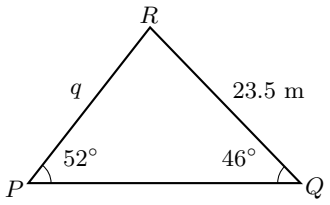
When two angles and any side are known (**AAS** or **ASA**), the sine law finds the other sides.

1. If the known side is not opposite a known angle, find the third angle first (180° minus the other two). Now there is a known pair.
2. Write $\dfrac{\text{unknown side}}{\sin(\text{its opposite angle})} = \dfrac{\text{known side}}{\sin(\text{its opposite angle})}$ and multiply to isolate the side.
3. Keep unrounded values in the calculator and round only the final answer.

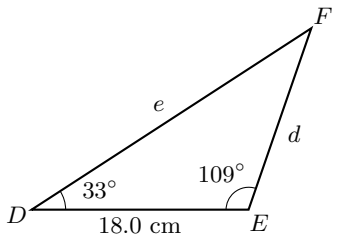
Example 5

Round sides to the nearest tenth.

(a) In $\triangle PQR$, $\angle P = 52°$, $\angle Q = 46°$ and $p = 23.5$ m. Determine q.



(b) In $\triangle DEF$, $\angle D = 33°$, $\angle E = 109°$ and $DE = 18.0$ cm. Determine d and e.



5. Finding an Angle

To find an angle, put the sines on top: $\dfrac{\sin B}{b} = \dfrac{\sin A}{a}$ gives $\sin B = \dfrac{b \sin A}{a}$, then $\angle B = \sin^{-1}\left(\dfrac{b \sin A}{a}\right)$ (TI-84: 2ND [SIN^{-1}]).

The inverse sine key only gives an angle from 0° to 90°. Since $\sin(180° - B) = \sin B$, an obtuse angle has the same sine, so we must know the angle is acute. In this lesson every question tells you why it is:

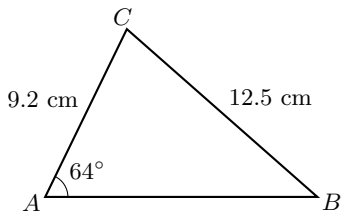
- the triangle is **acute** (all its angles are less than 90°), or
- the known angle is opposite the **longer** of the two given sides. The larger side is opposite the larger angle, so the unknown angle is smaller than the known angle; a triangle has at most one angle of 90° or more, so the unknown angle is acute.

When neither is true there may be two triangles: that is the **ambiguous case** (Lesson 7). **Solving a triangle** means finding all of its unknown sides and angles.

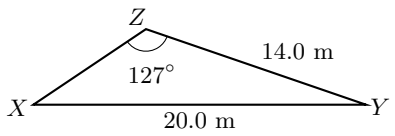
Example 6

Round sides to the nearest tenth and angles to the nearest degree.

(a) In $\triangle ABC$, $\angle A = 64°$, $a = 12.5$ cm and $b = 9.2$ cm. Determine $\angle B$, $\angle C$ and c.



(b) In $\triangle XYZ$, $\angle Z = 127°$, $z = 20.0$ m and $x = 14.0$ m. Solve the triangle.



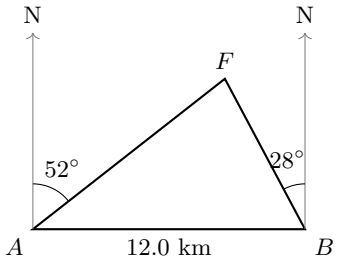
6. Problems in Context

Sketch the situation, label what is known, and look for a triangle with a known pair (or one you can complete with the angle sum). Some problems need a second step with SOH CAH TOA. Two kinds of angles appear often:

- A **bearing** such as N52°E means: face north, then turn 52° toward the east.
- An **angle of elevation** is measured up from the horizontal to the line of sight.

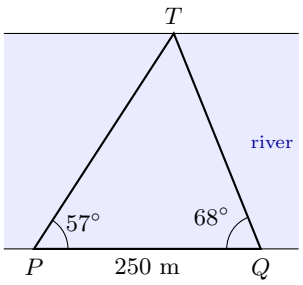
Example 7

Fire lookout tower B is 12.0 km due east of tower A. A fire F is spotted on a bearing of N52°E from A and N28°W from B. How far is the fire from each tower, to the nearest tenth of a kilometre?



Example 8

To find the width of a river, a surveyor marks points P and Q, 250 m apart, along one straight bank. A tree T stands on the far bank. She measures $\angle TPQ = 57°$ and $\angle TQP = 68°$.



(a) Determine PT to the nearest tenth of a metre.

(b) Determine the width of the river to the nearest tenth of a metre.

(c) Determine the area of $\triangle PQT$ to the nearest square metre.

Lesson 6: The Cosine Law

Finding the third side from two sides and the angle between them, and any angle from three sides.

1. When the Sine Law Cannot Start

The sine law needs a known pair (an angle and its opposite side). Two cases have no known pair:

- **SAS**: two sides and the **contained** angle (the angle between them);
- **SSS**: all three sides.

These triangles can be solved by splitting them into right triangles, or in one step with the cosine law.

Example 1

Decide whether the sine law can be used as the first step. If not, name the case (SAS or SSS).

(a) $\angle A$, $\angle C$, c _____

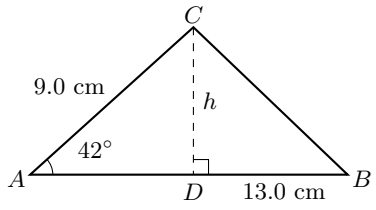
(b) b, c, $\angle A$ _____

(c) a, b, c _____

(d) $\angle B$, b, a $(b > a)$ _____

Example 2

In $\triangle ABC$, $\angle A = 42°$, $AC = 9.0$ cm and $AB = 13.0$ cm. The altitude CD has length h. Determine BC to the nearest tenth using right triangles only.

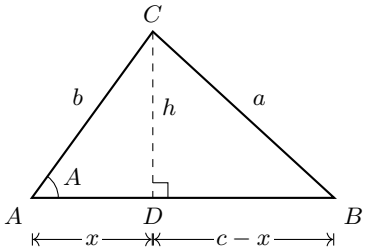


2. The Cosine Law

Repeating Example 2 with letters gives a formula. Put A at the left end of side c, and let $AD = x$, so $DB = c - x$.

Example 3

Complete the proof.



In $\triangle ACD$: $\cos A = \dfrac{x}{b}$, so $x =$ _____, and $h^2 + x^2 =$ _____.

In $\triangle BCD$: $a^2 = h^2 + (c - x)^2$

Expand: $a^2 = h^2 + c^2 - 2cx + x^2$

Regroup: $a^2 = (h^2 + x^2) + c^2 - 2cx$

Substitute: $a^2 =$ _____ $+ c^2 - 2c($_____$)$

So $a^2 = b^2 + c^2 - 2bc\cos A$

The Cosine Law. In any $\triangle ABC$: $a^2 = b^2 + c^2 - 2bc\cos A$ $\quad b^2 = a^2 + c^2 - 2ac\cos B$ $\quad c^2 = a^2 + b^2 - 2ab\cos C$

The side on the left is opposite the angle on the right; the other two sides are the ones that form that angle. If the angle is $90°$, then $\cos 90° = 0$ and the law becomes the Pythagorean theorem.

3. Finding a Side (SAS)

Substitute the two sides and the contained angle, then take the square root. Enter the whole expression at once, for example $\sqrt{9^2 + 13^2 - 2(9)(13)\cos 42°}$. Do **not** subtract $2bc$ from $b^2 + c^2$ before multiplying by $\cos A$: the multiplication comes first. When the angle is obtuse, $\cos A < 0$, so the last term is *added* and the third side is longer.

Example 4

Determine the third side. Round to the nearest tenth in (a) and (b); give an exact answer in (c).

(a) $\triangle ABC$ from Example 2: $\angle A = 42°$, $b = 9.0$ cm, $c = 13.0$ cm

(b) $\triangle PQR$: $\angle Q = 128°$, $p = 7.5$ m, $r = 11.2$ m

(c) $\triangle DEF$: $\angle E = 120°$, $d = 3$, $f = 5$ (no calculator)

4. Finding an Angle (SSS)

Solve the cosine law for the cosine of the angle.

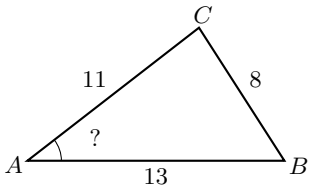
$$\cos A = \frac{b^2 + c^2 - a^2}{2bc} \qquad \cos B = \frac{a^2 + c^2 - b^2}{2ac} \qquad \cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

- The side *subtracted* in the numerator is the side opposite the angle you want.
- $\cos^{-1}$ gives an angle from $0°$ to $180°$. A **negative** cosine means the angle is **obtuse**. There is never a second answer, so the cosine law is the safe way to find an angle that might be obtuse.
- The **largest angle** is opposite the **longest side**, and the smallest angle is opposite the shortest side.

Example 5

Round angles to the nearest degree.

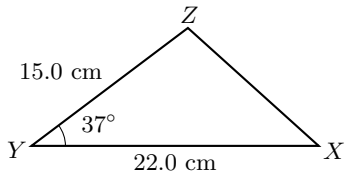
(a) Determine $\angle A$.



(b) A triangle has sides 6.5 cm, 9.2 cm and 13.8 cm. Determine its largest angle.

Example 6

Solve $\triangle XYZ$ if $\angle Y = 37°$, $x = 15.0$ cm and $z = 22.0$ cm. Round sides to the nearest tenth and angles to the nearest degree.

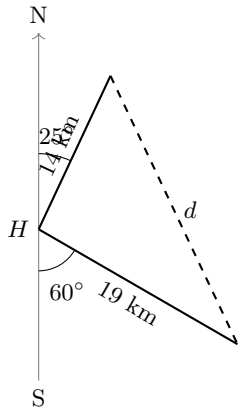


5. Problems in Context

Sketch the situation and look for two sides with the angle between them (SAS) or three sides (SSS).

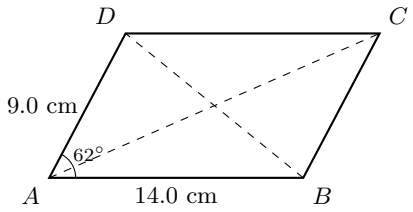
Example 7

Two boats leave a harbour H at the same time. One travels 14 km on a bearing of N25°E. The other travels 19 km on a bearing of S60°E. How far apart are the boats, to the nearest tenth of a kilometre?



Example 8

Parallelogram $ABCD$ has $AB = 14.0$ cm, $AD = 9.0$ cm and $\angle DAB = 62°$. Determine the lengths of both diagonals to the nearest tenth.



Lesson 7: The Ambiguous Case

Two sides and an angle that is not between them can give no triangle, one triangle or two triangles; how to tell which, and how to find every solution.

1. Which Measurements Fix a Triangle?

Three measurements, at least one of them a side, describe a triangle. With **SSS**, **SAS** (the angle is *between* the two sides), **ASA** and **AAS**, every triangle drawn from the measurements is congruent to every other: there is **exactly one** triangle.

With **SSA** (two sides and an angle that is *not* between them) this is not guaranteed. There may be no triangle, one triangle or two different triangles. SSA is called the **ambiguous case**.

The sine law shows where the trouble comes from. It gives the *sine* of the unknown angle, and two angles between 0° and 180° have the same sine (sine is positive in quadrants I and II). The calculator's $\sin^{-1}$ key gives only the acute one.

For $0° < \theta < 180°$: $\sin\theta = \sin(180° - \theta)$.
If $\sin B = k$ with $0 < k < 1$, then $B = \sin^{-1} k$ (acute) **or** $B = 180° - \sin^{-1} k$ (obtuse). Each one gives a triangle only if the angles still add to less than 180°.

Example 1

For each $\triangle PQR$, name the type of information given (SSS, SAS, ASA, AAS or SSA) and state whether it fixes exactly one triangle.

(a) $p = 7$ cm, $q = 9$ cm, $r = 12$ cm

(b) $\angle P = 50°$, $q = 8$ cm, $r = 11$ cm

(c) $\angle P = 50°$, $\angle Q = 72°$, $r = 10$ cm

(d) $\angle P = 50°$, $p = 7$ cm, $q = 8$ cm

Example 2

Two students each sketched $\triangle ABC$ with $\angle A = 40°$, $AC = 10$ cm and $BC = 7$ cm. Mia's triangle has an acute angle at B. Theo's has an obtuse angle at B.

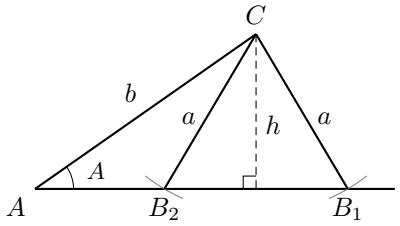
(a) Use the sine law to find both possible measures of $\angle B$, to the nearest tenth of a degree.

(b) Explain how both students can be right.

2. Picturing SSA: The Swinging Side

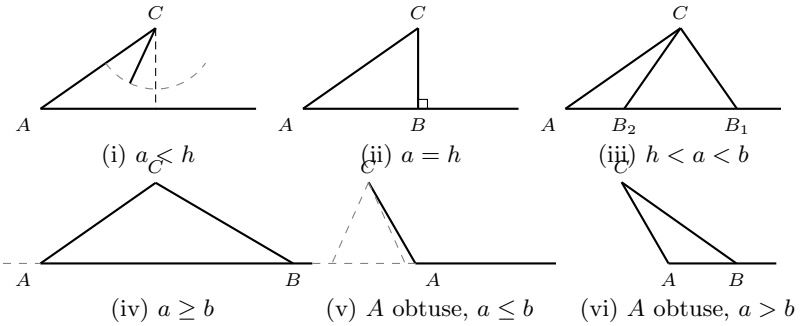
Suppose $\angle A$, side $b = AC$ and side $a = BC$ (opposite $\angle A$) are given. Draw $\angle A$ and mark C on one arm, b units from A. Side a hangs from C and swings like a compass; B is wherever it lands on the other arm. The shortest distance from C to that arm is the **height** h. From the right triangle, $\sin A = \dfrac{h}{b}$.

$h = b \sin A$ Compare a (the side opposite the given angle) with h and with b.



Example 3

Complete the summary. Sketches (i)–(iv) have an acute given angle A; sketches (v)–(vi) have an obtuse given angle A.



Given angle A	Condition	Number of triangles
acute	$a < h$	
acute	$a = h$	
acute	$h < a < b$	
acute	$a \geq b$	
obtuse or right	$a \leq b$	
obtuse or right	$a > b$	

Quick check: if the side opposite the given angle is the *longer* of the two given sides ($a \geq b$), there is exactly one triangle (or none if $a = b$ with A obtuse or right). Only an acute A with $a < b$ can give two.

Example 4

In $\triangle ABC$, $\angle A = 30°$ and $b = 14$ cm. Determine the height h from C, then the number of triangles for each length of a.

$h =$ _____	$a = 5$ cm	$a = 7$ cm	$a = 10$ cm	$a = 16$ cm
Number of triangles				

For an obtuse angle, $\angle A = 115°$ and $b = 8$ cm: $a = 6$ cm gives _____ $a = 11$ cm gives _____

3. Solving the Ambiguous Case

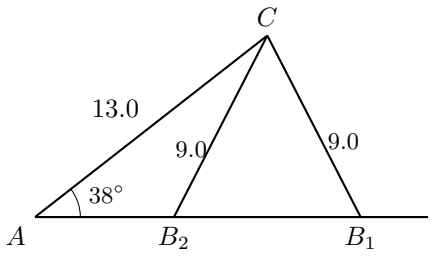
Solving SSA (given $\angle A$, a and b; find $\angle B$)

1. Sketch. Optionally count the triangles first with $h = b \sin A$.
2. Sine law: $\sin B = \dfrac{b \sin A}{a}$. If this is greater than 1, there is no triangle.
3. $B_1 = \sin^{-1}(\sin B)$ (acute) and $B_2 = 180° - B_1$ (obtuse).
4. Keep B_2 only if $A + B_2 < 180°$.
5. Finish each triangle: $C = 180° - A - B$, then the third side with the sine law.

Keep unrounded values in the calculator until the final answers.

Example 5

Solve all possible triangles ABC in which $\angle A = 38°$, $a = 9.0$ cm and $b = 13.0$ cm. Round angles to the nearest degree and sides to the nearest tenth.



Example 6

Determine all possible measures of the indicated angle(s), to the nearest degree. If there is no triangle, explain why.

(a) $\triangle PQR$: $\angle P = 47°$, $p = 15$ m, $q = 11$ m. Find $\angle Q$ and $\angle R$.

(b) $\triangle XYZ$: $\angle X = 62°$, $x = 8$ cm, $y = 10$ cm. Find $\angle Y$.

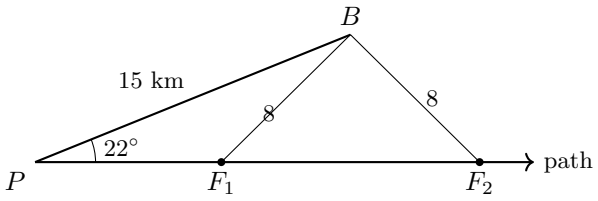
(c) $\triangle KLM$: $\angle K = 108°$, $k = 20$ cm, $l = 14$ cm. Find $\angle L$, $\angle M$ and side m (nearest tenth).

4. Problems with Two Possible Positions

In a word problem the ambiguous case shows up when a fixed length swings from a fixed point: a gate on a hinge, a strut, a rope, or the edge of a range (radio signal, sprinkler, sound). Sketch both positions, solve both triangles, then decide whether each one makes sense.

Example 7

A ferry leaves port P and travels in a straight line. A buoy B is 15 km from P, and the angle between the ferry's path and the line PB is 22°. The buoy's radio beacon reaches 8 km. At what distances from P along its path is the ferry exactly 8 km from the buoy? Round to the nearest tenth of a kilometre.



Example 8

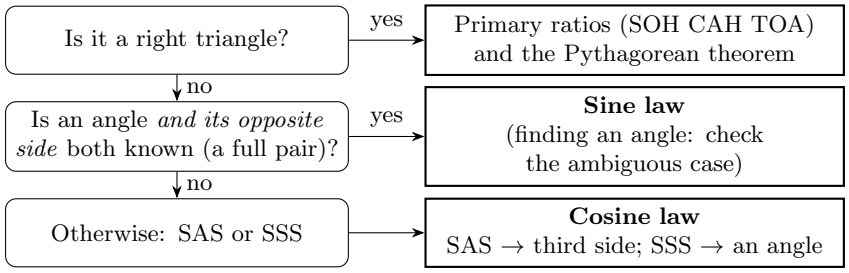
A roof rafter AC is 6.0 m long and makes an angle of 28° with the horizontal ceiling beam at A. A 3.5 m support strut is hinged at the top end C of the rafter and swings down to rest on the beam. How far from A can the foot of the strut be? Round to the nearest tenth of a metre.

Lesson 8: Choosing the Sine Law or the Cosine Law

Deciding which tool fits a triangle, then using primary ratios, the sine law and the cosine law together in bearings, circles, multi-triangle problems and areas.

1. Choosing a Method

Before solving, look at what is known in the triangle that contains the unknown.



With only the three angles (AAA) the triangle's shape is known but not its size: no side can be found. After the cosine law has given one missing part, a full pair is known, so the sine law (easier) usually finishes the triangle. To find an angle, the cosine law is safer: it is never ambiguous, and a negative cosine shows an obtuse angle.

Example 1

For each, name the most appropriate method and justify the choice. Do not solve.

(a) $\triangle ABC$: $\angle C = 90°$, $\angle A = 34°$, $c = 12$; find b

(b) $\triangle DEF$: $d = 7$, $e = 9$, $\angle F = 52°$; find f

(c) $\triangle GHJ$: $\angle G = 41°$, $g = 10$, $h = 13$; find $\angle H$

(d) $\triangle KLM$: $k = 5$, $l = 8$, $m = 11$; find the largest angle

(e) $\triangle PQR$: $\angle P = 50°$, $\angle Q = 60°$, $\angle R = 70°$; find p

(f) $\triangle STU$: $\angle S = 35°$, $\angle T = 105°$, $s = 8$; find u

Example 2

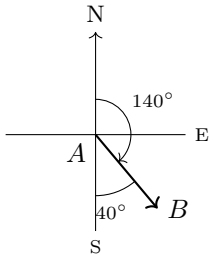
The minute hand of a clock is 9 cm long and the hour hand is 6 cm long. The 12 hour marks are $360° \div 12 = 30°$ apart, and the hour hand moves $0.5°$ each minute (halfway between marks at the half hour).

(a) Find the angle between the hands at 4:00, and the distance between their tips to the nearest tenth.

(b) Repeat for 2:30.

2. Bearings and Directions

A **bearing** gives a direction as an angle measured **clockwise from north**, written with three digits: east is 090°, south is 180°, west is 270°. The bearing of B from A is measured at A.
A direction can also be written from north or south toward east or west: **S** 40° **E** means "face south, turn 40° toward east" (the same as 140°).
The **back bearing** (the bearing of A from B) differs by 180°: if B is on a bearing of 065° from A, then A is on a bearing of 245° from B. North lines at different points are parallel, so alternate and co-interior angles give the angles inside a triangle.



bearing 140° = S 40° E

Example 3

(a) 065° = _____ (b) 250° = _____ (c) 310° = _____ (d) 165° = _____

(e) S 15° W = _____ (f) N 20° W = _____ (g) S 62° E = _____

(h) B is on a bearing of 300° from A. The bearing of A from B is _____.

Example 4

Two aircraft leave an airport at the same time. One flies on a bearing of 050° at 540 km/h; the other flies on a bearing

of 165° at 480 km/h. How far apart are they after 15 minutes, to the nearest kilometre?

Example 5

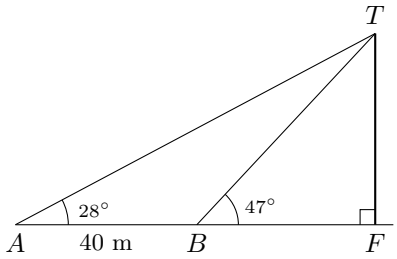
A ship sails 18 km from port P on a bearing of 060° to Q, then 25 km on a bearing of 170° to R. Sketch the route. Find the distance and the bearing the ship must sail to return directly from R to P (nearest tenth of a km and nearest degree).

3. Problems with Two Triangles

When the unknown is in a triangle that does not have enough information, solve a neighbouring triangle first to get a shared side or angle. Mark every known value on the sketch and look for supplementary angles along straight lines (180°) and right angles.

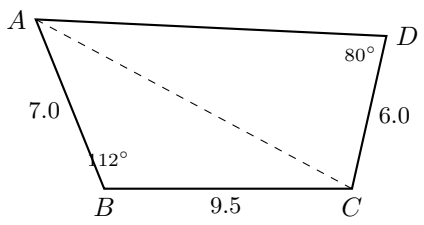
Example 6

From point A on level ground, the angle of elevation to the top T of a radio mast is 28°. After walking 40 m straight toward the mast to point B, the angle of elevation is 47°. Find the height of the mast, to the nearest tenth of a metre.



Example 7

A garden plot $ABCD$ is split by the diagonal path AC. $AB = 7.0$ m, $BC = 9.5$ m, $\angle ABC = 112°$, $CD = 6.0$ m and $\angle ADC = 80°$. Find the length of AC, the length of AD and the perimeter of the plot, to the nearest tenth of a metre.



4. Triangles in Circles

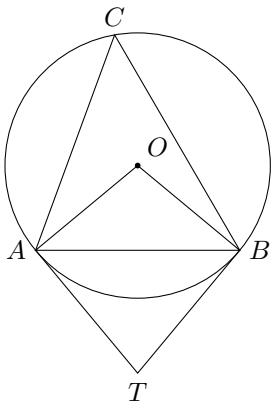
Radii of the same circle are equal, so a chord and two radii form an **isosceles** triangle. Useful facts from earlier courses:

- The perpendicular from the centre to a chord **bisects** the chord.
- The **central angle** is **twice** the **inscribed angle** on the same arc. Inscribed angles on the same arc are **equal**. (An angle inscribed in a semicircle is 90°.)
- A **tangent** is **perpendicular** to the radius at the point of tangency.

Example 8

A circle with centre O has radius 6.5 cm. Chord AB is 10.0 cm long, and C is a point on the major arc. The tangents at A and B meet at T. Round angles to the nearest degree and lengths to the nearest tenth.

(a) Find $\angle AOB$. (b) Find $\angle ACB$. (c) Find the distance from O to the chord. (d) Find $\angle ATB$ and the length TA.



5. Area and Perimeter of a Triangle

In $\triangle ABC$, the height from B to side b is $a\sin C$, so Area $= \frac{1}{2}$(base)(height) $= \frac{1}{2}ab\sin C$. This uses two sides and the angle *between* them (SAS); for SSS, find an angle with the cosine law first.

$$\text{Area} = \frac{1}{2}ab\sin C = \frac{1}{2}bc\sin A = \frac{1}{2}ac\sin B \quad \text{(two sides and the contained angle)}$$

Example 9

(a) A triangular lot has two sides of 42 m and 55 m with an angle of 74° between them. Find its area (nearest square metre) and its perimeter (nearest tenth of a metre).

(b) Find the area of a triangle with sides 7 cm, 9 cm and 12 cm, to the nearest tenth.

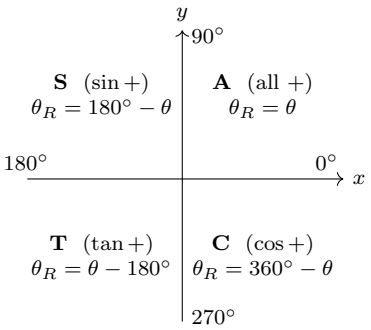
(c) Find the area of the garden plot in Example 7, to the nearest tenth.

Lesson 9: Unit 3 Review

Every idea of the unit on one page, mixed examples that combine them, and a review assignment written like the unit test.

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1. Unit Summary



Standard position: vertex at the origin, **initial arm** on the positive x-axis, rotation counterclockwise to the **terminal arm**. The **reference angle** θ_R is the acute angle between the terminal arm and the x-axis. A **quadrantal angle** $(0°, 90°, 180°, 270°)$ lies on an axis.

Point $P(x, y)$ on the terminal arm: $r = \sqrt{x^2 + y^2}$,

$$\sin\theta = \frac{y}{r}, \qquad \cos\theta = \frac{x}{r}, \qquad \tan\theta = \frac{y}{x} \quad (x \neq 0).$$

Reflecting $P(x, y)$ in the y-axis, the origin and the x-axis gives the angles $180° - \theta$, $180° + \theta$ and $360° - \theta$ (same θ_R).

Exact values (special triangles $1, 1, \sqrt{2}$ and $1, \sqrt{3}, 2$)

θ	$0°$	$30°$	$45°$	$60°$	$90°$	$180°$	$270°$
$\sin\theta$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1	0	-1
$\cos\theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0	-1	0
$\tan\theta$	0	$\frac{\sqrt{3}}{3}$	1	$\sqrt{3}$	undef.	0	undef.

Solving $\sin\theta = a$, $\cos\theta = a$, $\tan\theta = a$ for $0° \le \theta < 360°$:

1. Find θ_R: the inverse key with $|a|$ (or a special triangle).
2. Use the sign of a and CAST to pick the two quadrants.
3. Place θ_R in each quadrant.

$|a| > 1$ for sin or cos: no solution. ± 1 or 0: quadrantal answers.

Sine law $\dfrac{a}{\sin A} = \dfrac{b}{\sin B} = \dfrac{c}{\sin C}$

Cosine law $a^2 = b^2 + c^2 - 2bc\cos A, \quad \cos A = \dfrac{b^2 + c^2 - a^2}{2bc}$

Area $\text{Area} = \dfrac{1}{2}ab\sin C$ (two sides and the angle between them)

Ambiguous case (SSA): given $\angle A$, a and b; let $h = b\sin A$.

$\angle A$ acute	
$a < h$	no triangle
$a = h$	one (right) triangle
$h < a < b$	two triangles: B and $180° - B$
$a \ge b$	one triangle
$\angle A$ obtuse: $a \le b$ none, $a > b$ one	

Choosing. Right triangle: SOH CAH TOA. A side and its opposite angle known (AAS, ASA after finding the third angle, SSA): sine law. Otherwise (SAS, SSS): cosine law. A negative cosine means an obtuse angle.

Good habits. Calculator in **Degree mode**. Sketch every problem and label sides with the lowercase letter of the opposite angle. Keep full calculator values between steps and round only the final answer. A **bearing** is measured clockwise from north, written with three digits (e.g. 062°). Common errors: θ_R is measured to the x-axis, never the y-axis; the inverse key gives only one angle; in the cosine law, do the multiplication $2bc\cos A$ before subtracting.

2. Angles, Points and Ratios

Example 1

(a) State the quadrant and the reference angle of each angle.

θ	73°	164°	215°	342°
Quadrant				
θ_R				

(b) Determine the four angles from 0° to 360° that have a reference angle of 19°. _____

(c) The point $P(3, 7)$ is on the terminal arm of angle θ. Determine θ to the nearest tenth of a degree. Then give the angles whose terminal arms pass through the reflections $Q(-3, 7)$, $R(-3, -7)$ and $S(3, -7)$.

Example 2

Each point is on the terminal arm of angle θ, $0° \leq \theta < 360°$. Determine the exact values of $\sin\theta$, $\cos\theta$ and $\tan\theta$, then θ to the nearest tenth of a degree.

(a) $P(-8, -15)$ (b) $P(4, -6)$ (c) $P(-5, 0)$

Example 3

Given $\sin\theta = -\frac{2}{5}$ and $\tan\theta > 0$, determine the exact values of $\cos\theta$ and $\tan\theta$.

3. Exact Values and Solving Equations

Example 4

No calculator.

(a) Determine the exact value.

(i) $\sin 135°$ (ii) $\cos 210°$ (iii) $\tan 300°$ (iv) $\cos 270°$

(b) Evaluate exactly: $2\sin 150° - \tan^2 225° + \cos 180°$

(c) Solve exactly for $0° \leq \theta < 360°$.

(i) $\cos\theta = -\dfrac{\sqrt{3}}{2}$ (ii) $\tan\theta = -1$ (iii) $2\sin\theta + \sqrt{2} = 0$

Example 5

Solve for $0° \leq \theta < 360°$, to the nearest tenth of a degree.

(a) $\sin\theta = 0.38$ (b) $\tan\theta = -2.5$ (c) $3\cos\theta + 2 = 0$

(d) $\sin\theta = -1.3$ (e) $\cos\theta = -1$

4. The Sine Law, the Cosine Law and the Ambiguous Case

Example 6

Sides to the nearest tenth, angles to the nearest degree.

(a) In $\triangle PQR$, $\angle P = 48°$, $\angle Q = 71°$ and $p = 15.2$ cm. Determine r.

(b) A triangle has sides 7 m, 9 m and 13 m. Determine its largest angle.

(c) Determine the area of the triangle in (b) to the nearest square metre.

Example 7

In $\triangle ABC$, $\angle A = 38°$, $a = 9$ cm and $b = 12$ cm.

(a) Show that two triangles are possible.

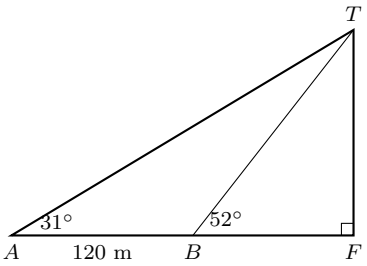
(b) Solve both triangles (sides to the nearest tenth, angles to the nearest degree).

(c) How many triangles are possible if instead (i) $a = 7$ cm, (ii) $a = 13$ cm, (iii) $a = 12 \sin 38°$ cm?

5. Problem Solving

Example 8

(a) Two points A and B on level ground are 120 m apart and in line with the base F of a building. From A the angle of elevation of the top T is 31°; from B, closer to the building, it is 52°. Determine the height of the building to the nearest tenth of a metre.



(b) Two boats leave a dock at the same time. One travels at 14 km/h on a bearing of 030°; the other travels at 18 km/h on a bearing of 145°. Sketch a diagram and determine how far apart they are after 1.5 h, to the nearest tenth of a kilometre.